



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

ITEC 204 – Data Structures and Algorithms

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- I. **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer: The main objective of this laboratory exercise is to practice basic list operations like adding, searching, and removing data by making a basic ticket management system. By building this program, we can see how linear data structures are actually applied in real-world IT scenarios, rather than just learning the theory.

- II. **Program Operations:** Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident: Uses the append() method to insert a new ticket dictionary at the end of the list.

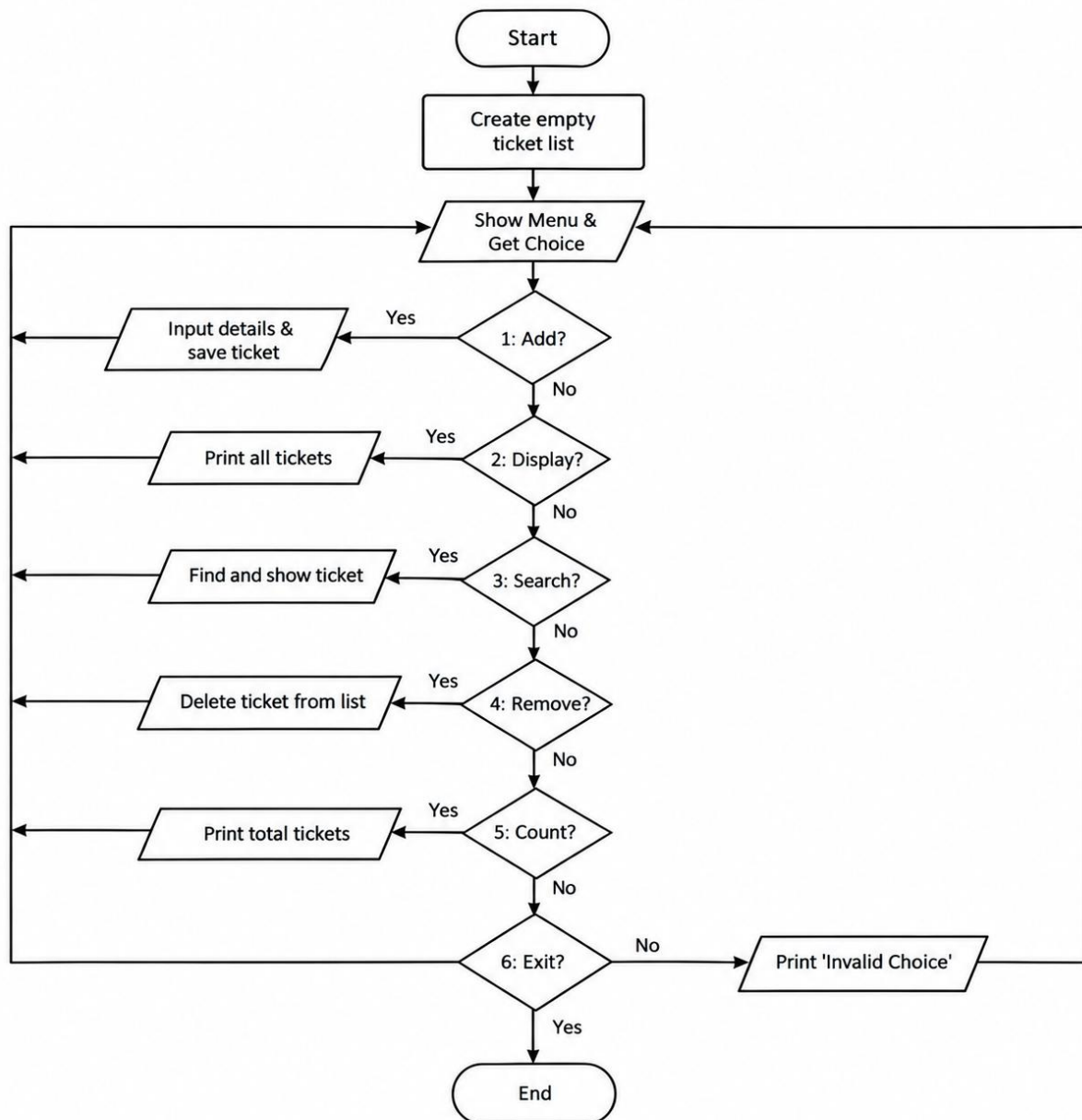
Display Incidents: Iterates through the list using a for loop to print all active tickets.

Search Incident: Performs a linear search through the list, matching the provided Incident ID against the existing records.

Remove Resolved Incident: Uses the pop() method to remove a specific ticket from the list once its Incident ID is found.

Count Active Incidents: Uses Python's built in len() function to return the current number of elements in the list.

- III. **Flowchart:** Create a flowchart showing the overall process and logic of your program.



IV. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.

This laboratory exercise demonstrated how linear data structures, lists, can be used to manage records sequentially. By implementing basic operations such as appending, searching, and removing elements, I gained a practical understanding of how data structures form the skelato of simple management software.



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Scenario: IT support teams frequently deal with automated incident tickets generated by various system bots. This program acts as an IT Automation Incident Ticket Manager, allowing support staff to add, view, search, remove, and count active incident tickets efficiently to ensure no automated alerts are missed.

Data Structure Used: The program utilizes a **List** as its primary linear data structure. A list was chosen because it maintains the sequential order of added tickets, allows for dynamic resizing as tickets are added or removed, and supports linear traversal which is necessary for searching and displaying all records. Each element inside the list is a dictionary holding the ticket's specific attributes (id, bot, and description).

```
Source Code: active_tickets = []

while True:
    print("\n=====")
    print(" IT Automation Incident Ticket Manager ")
    print("=====")
    print("1. Add a new incident ticket")
    print("2. Display all active incident tickets")
    print("3. Search for a specific incident ticket")
    print("4. Remove a resolved incident ticket")
    print("5. Display total number of active tickets")
    print("6. Exit")
    print("=====")

    choice = input("Enter your choice (1-6): ")

    if choice == '1':
        inc_id = input("Enter Incident ID: ")
        bot = input("Enter Bot Name: ")
        desc = input("Enter Short Description: ")

        ticket = {"id": inc_id, "bot": bot, "description": desc}
        active_tickets.append(ticket)
        print("Ticket added successfully!")

    elif choice == '2':
        if len(active_tickets) == 0:
            print("No active tickets right now.")
        else:
            print("\n--- Active Tickets ---")
            for i in range(len(active_tickets)):
                t = active_tickets[i]
                print(f"Ticket {i+1} | ID: {t['id']} | Bot: {t['bot']} | Desc: {t['description']}")
```



```
elif choice == '3':
    search_id = input("Enter Incident ID to search: ")
    found = False
    for t in active_tickets:
        if t["id"] == search_id:
            print("\nTicket Found!")
            print("Incident ID:", t["id"])
            print("Bot:", t["bot"])
            print("Description:", t["description"])
            found = True
            break

    if found == False:
        print("Ticket not found.")

elif choice == '4':
    remove_id = input("Enter Incident ID to remove: ")
    found = False
    for i in range(len(active_tickets)):
        if active_tickets[i]["id"] == remove_id:
            active_tickets.pop(i)
            print("Ticket removed successfully!")
            found = True
            break

    if found == False:
        print("Ticket not found. Cannot remove.")

elif choice == '5':
    print("Total active tickets:", len(active_tickets))

elif choice == '6':
    print("Exiting program...")
    break

else:
    print("Invalid choice. Please try again.")
```



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Program Output:

Figure 1: Adding a new incident ticket

```
PS C:\Users\PC\Downloads\groupact1> & C:\Users\PC\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/PC/Downloads/groupact1/practicefolder/lab_exer_1.py

=====
IT Automation Incident Ticket Manager
=====
1. Add a new incident ticket
2. Display all active incident tickets
3. Search for a specific incident ticket
4. Remove a resolved incident ticket
5. Display total number of active tickets
6. Exit
=====
Enter your choice (1-6): 1
Enter Incident ID: INC1392939
Enter Bot Name: BOT-Inventory
Enter Short Description: Failed to generate the daily report
Ticket added successfully!
```

Figure 2: Displaying all active tickets

```
=====
IT Automation Incident Ticket Manager
=====
1. Add a new incident ticket
2. Display all active incident tickets
3. Search for a specific incident ticket
4. Remove a resolved incident ticket
5. Display total number of active tickets
6. Exit
=====
Enter your choice (1-6): 2

--- Active Tickets ---
Ticket 1 | ID: INC1392939 | Bot: BOT-Inventory | Desc: Failed to generate the daily report
Ticket 2 | ID: INC1392940 | Bot: BOT-Email | Desc: Failed to send the scheduled notification
Ticket 3 | ID: INC1392941 | Bot: BOT-DataSync | Desc: Encountered an error during data transfer
Ticket 4 | ID: INC1392942 | Bot: BOT-Invoice | Desc: Failed to process an invoice
Ticket 5 | ID: INC1392943 | Bot: BOT-Report | Desc: Failed to generate the weekly report
Ticket 6 | ID: INC1392944 | Bot: BOT-FileTransfer | Desc: Failed to upload the required file
Ticket 7 | ID: INC1392945 | Bot: BOT-DataEntry | Desc: Encountered an error while entering records
Ticket 8 | ID: INC1392946 | Bot: BOT-Backup | Desc: Failed to complete the scheduled backup
Ticket 9 | ID: INC1392947 | Bot: BOT-Validation | Desc: Failed to validate the submitted records
Ticket 10 | ID: INC1392948 | Bot: BOT-Notification | Desc: Failed to send the system alert
```

Figure 3: Searching for a specific ticket

```
=====
IT Automation Incident Ticket Manager
=====
1. Add a new incident ticket
2. Display all active incident tickets
3. Search for a specific incident ticket
4. Remove a resolved incident ticket
5. Display total number of active tickets
6. Exit
=====
Enter your choice (1-6): 3
Enter Incident ID to search: INC1392939

Ticket Found!
Incident ID: INC1392939
Bot: BOT-Inventory
Description: Failed to generate the daily report
```



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Figure 4: Removing a resolved ticket

```
=====
IT Automation Incident Ticket Manager
=====
1. Add a new incident ticket
2. Display all active incident tickets
3. Search for a specific incident ticket
4. Remove a resolved incident ticket
5. Display total number of active tickets
6. Exit
=====
Enter your choice (1-6): 4
Enter Incident ID to remove: INC1392939
Ticket removed successfully!
```

Figure 5: Displaying the total ticket count

```
=====
IT Automation Incident Ticket Manager
=====
1. Add a new incident ticket
2. Display all active incident tickets
3. Search for a specific incident ticket
4. Remove a resolved incident ticket
5. Display total number of active tickets
6. Exit
=====
Enter your choice (1-6): 5
Total active tickets: 9
=====
```